THE SPECTRUM EDUCATION CONSULTANCY

P.L.E Spot-Check 6 (Indigo Episode)

MATHEMATICS

2024

Time Allowed: 2 hours 15 minutes

Index number									

Name: _____**Stream**___

School_____

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

Read the following instructions carefully.

1. This paper is made up of two Sections: A and B
2. Answers to both sections must be written in the spaces provided in full sentences.
3. Section A has 20 questions (40 marks)
4. Section B has 12 questions (60 marks)
5. Attempt all questions.
6. All answers must be written in blue or black ballpoint pen or ink but not in pencil. All work done in pencil, except diagrams, will **NOT** be marked.
7. Crossing out of answers will lead to loss of marks.
8. Any handwriting that cannot be easily read may lead to loss of marks

Spectrum

FOR EXAMINER'S USE ONLY		
QN. NO.	MARK	SIGN
1 - 10		
11 - 20		
21 - 30		
31 – 32		
33 – 34		
35 – 36		
37 – 38		
39 – 40		
41 – 42		
43 – 44		
45 – 46		
47 – 48		
49 – 50		
51 - 52		
53 – 55		
TOTAL		

SECTION A (40 Marks)

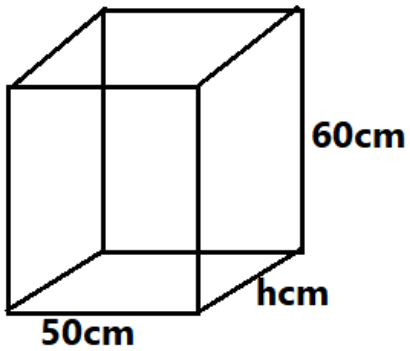
1. Add: 1056 + 05

2. Simplify: $^{+}10 - (^{-}2)$

3. Write 2204 in words.

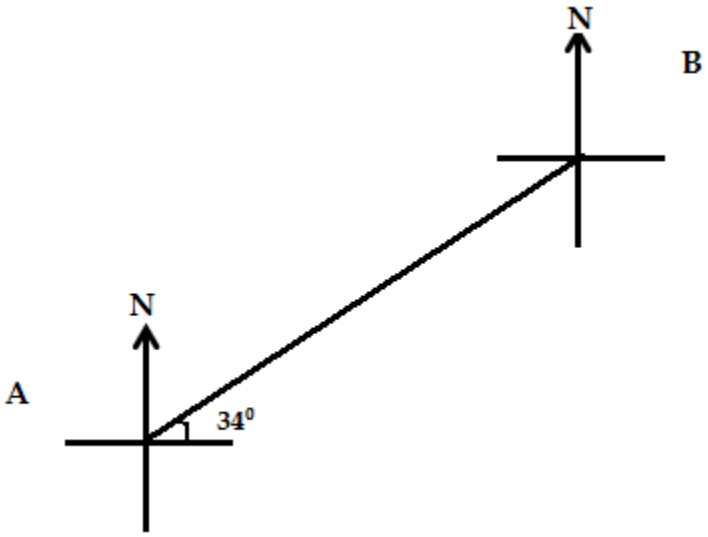
4. Cooking oil is sold in 250ml sachets. How much money would one pay for one litre of cooking oil if the market price of ½ a litre is sh. 3,750?

5. The volume of the tank below is 120,000cm^3. Find the height of the tan



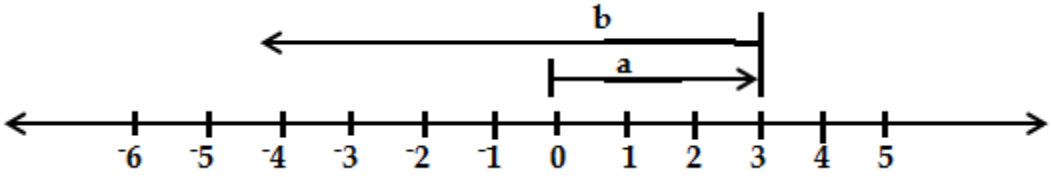
6. Eryama banked sh.600,000 in Equity bank. If the interest rate was 8% per annum. How much did he earn after six months?

7. Use the sketch diagram below to find the bearing of town **B** from **A**.



8. Multiply: 101_{two} x 11_{two}

9. Write the integers represented by the arrows **a** and **b** on the number line below.



a_____b_____

10. A car travels at an average speed of 90km/hr in 20 minutes. Find the distance covered by the car.

11. Find the next number in the sequence.
81, 64, 49, 36, _____

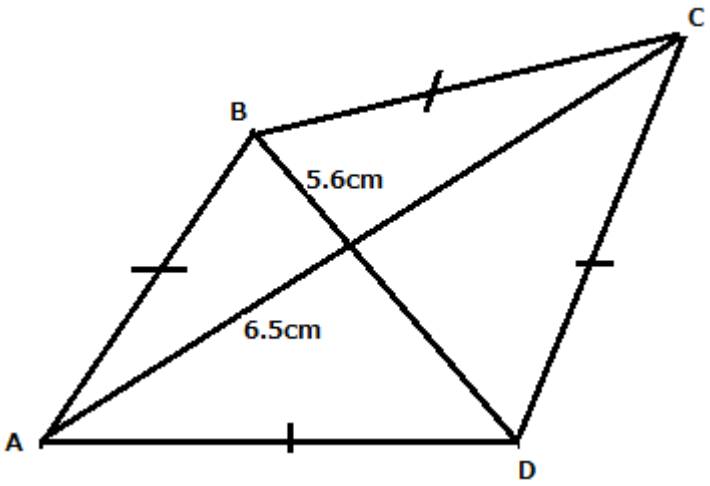
12. Round off 64.98 to the nearest tenth.

13. A wheel of diameter 28cm is rolling along a path of 176m.
How many complete resolutions does it make? (Use $\pi=\frac{22}{7}$)

14. Simplify: $\frac{0.48\ x\ 0.14}{0.6}$

15. The base area of a cylindrical tank is 1386cm^2. Find its capacity if it has a height of 100cm.

16. Below is a rhombus AB=BC=AD=DC and BD = 5.6cm, calculate its area.



17. The table below shows marks shored by some pupils in a primary seven class. Use it to find the median mark of the class.

No. of pupils	2	3	1	2
Marks scored	30	75	95	55

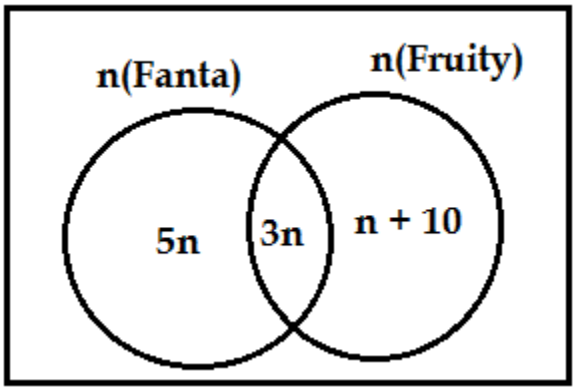
18. Express 0.3636----as a common fraction in its simplest form.

19. In a class, the ratio of boys to girls is 3:7 respectively. If a prefect is chosen at random from the class, what is the probability that the prefect will be a boy?

20. Find the number of elements in a set with 31 proper subsets.

SECTION B (60 marks)

21. The Venn diagram below shows different brands of soda taken by different people at a certain party. Study and use it to answer the questions that follow.



(a) Find the value of n. *(02 marks)*

(b) Find the total number of people who attended the party. (02 marks)

(c) How many people did not take fruity? *(02 marks)*

22. (a) Using a ruler, a pencil and a pair of compasses only, construct a kite STUV which ST=SV=5.5cm, SU=22cm and TV=6cm. *(04 marks)*

(b) Measure length TU_____cm *(01 mark)*

23. (a) Solve the equation: $2 - \frac{2x}{3} = 4$ *(02 marks)*

(b) Simplify: $\frac{2}{3}(6w - 3) - \frac{1}{2}(4 - 2w)$ *(05 marks)*

24. Kijambu went with a fifty thousand shilling note to the nearby shop and bought the following items.

3kg of sugar at sh.4000 per kg.

1 ½ kg of salt at sh.1200 per kg.

4 bars of soap for sh.5,000.

300ml of cooking oil at sh.6,000 per litre.

(a) If he was given a 10% discount, how much did he pay for all the items? *(05 marks)*

(b) How much in Kenyan shillings did Kijambu pay after the discount if 1ksh = ugsh.30? *(02 marks)*

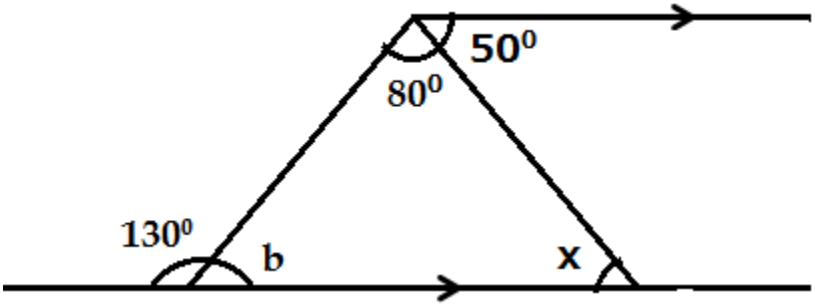
25. Use the equation $y = -2x + 4$ to complete the table below.

x	-3	-2	-1	0	1	_____
y	10	_____	_____	4	_____	0

(01 mark@)

26. (a) If 2y and 3y are complementary angles.
Find the value of y. *(02 marks)*

(b) In the figure below, find the size of angles **b** and **x**.



(i) Angle **b** (ii) angle **x**. *(03 marks)*

27. Square tiles of side 30cm were fixed on a floor measuring 9cm by 6cm.

(a) How many tiles were used? *(02 marks)*

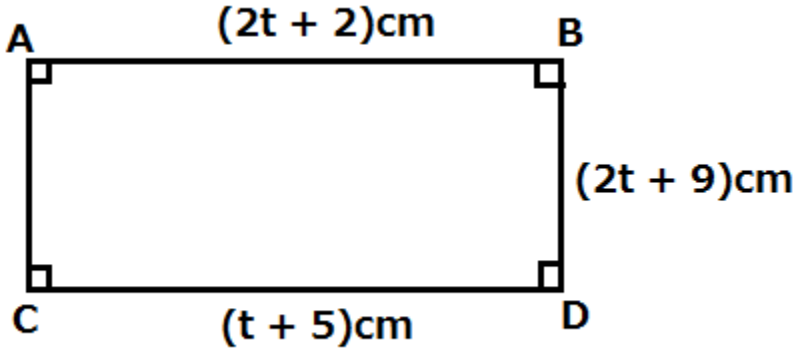
(b) If a box containing 50 tiles costs sh.250,000. How much was spent on the tiles? *(02 marks)*

28. Katumba has to fence around his rectangular plot of land which is 225cm by 165cm with poles placed 3m apart.

(a) How many poles will he need? *(02 marks)*

(b) If each pole costs sh.2000. How much money will he spend on fencing her plot of land? *(02 marks)*

29. Below is rectangle ABCD. Use it to answer questions that follow.

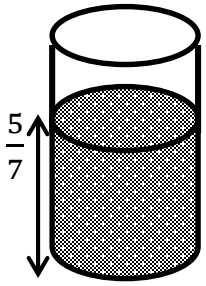


(a) Find the value of **t**? *(02 marks)*

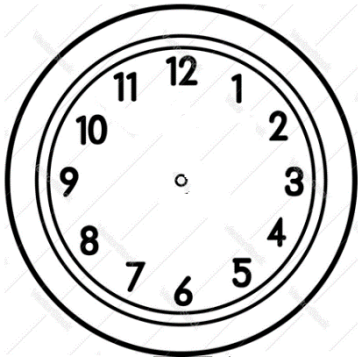
(b) Find the length of diagonal BD.

30. (a) After spending 30% of his salary, Okello was left with sh.210,000 only. How much did he spend?

(b) The Tank below was $\frac{5}{7}$ full of paraffin. When 220 litres of paraffin were drawn, it became $\frac{2}{5}$ full. Find the capacity of the tank.

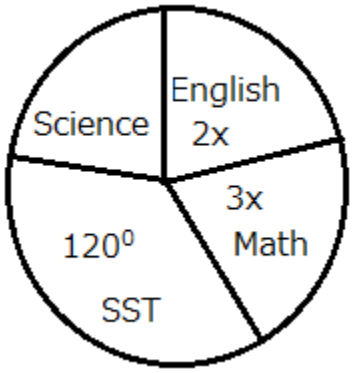


31. (a) On the clock face below, show twelve minutes to nine o'clock.



(b) Primary seven pupils went to bed at 9:30 p.m and woke up at 5:15 am. For how long were they in bed?

32. The pie chart below shows the quantity of books of various subjects in Apac primary school library.



(a) If there are 135 science books, find the number of English books in the library.

(b) Express the number of SST books as a percentage of all books in the library.

END

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